

Consumer-Relative Distinguishability in the Thermodynamic Limit

Andrew H. Bond*

San José State University, San José, California, USA

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The quantum results of this campaign were measured in finite-dimensional models, where every algebra is a full matrix algebra, every state has a density matrix, and every consumer has a partial trace. The local algebras of quantum field theory are not such objects. They are type III factors, which admit no trace, no density matrices, no partial traces, and no entanglement entropy of a region. This paper reports a measured bridge between the two settings. The consumer-relative quantities that carry every sealed claim of the campaign have well-defined thermodynamic limits, while the absolute quantities that do not survive the passage were never load-bearing. Three results support this. The type-III-native computational routes, through the relative modular operator and through states treated as expectation functionals, reproduce the sealed finite instrument at machine precision, six decades inside the declared agreement bar. A ladder of growing spin chains shows every consumer-relative restricted relative entropy converging exponentially, while a deliberately included absolute quantity, the entanglement entropy of the cut, diverges exactly extensively. The task-versus-fidelity flip of the earlier quantum paper persists at every rung of the ladder at fixed absolute budget, with its held-out task gap constant to nine decimals. A corollary is that the sealed quantum results of the campaign, computed at a single finite size, were already converged to their quoted precision and are effectively infinite-volume statements. Every number here is measured in finite models standing in for split inclusions, no claim is made that any extrapolated limit equals a field-theory value, and the consumer does not create anything physical by choosing what to observe.

I. INTRODUCTION

An earlier paper of this campaign defined consumer-relative distinguishability, the divergence between two quantum states as seen through the declared channel, task, and budget of a particular agent, and measured its behavior in small spin-chain models where every quantity is computable. Its three principal findings were a consumer hierarchy that divides one distinguishability total differently for different excitations, a strict task-versus-fidelity trade at scarce storage budgets called the flip, and a consumer-family consistency requirement that does constraining work exactly when the consumers share an event. Every one of those measurements lived inside finite-dimensional type I factors. Every algebra was a full matrix algebra, every state had a density matrix, and every consumer channel was built from a partial trace.

The observable algebra of a bounded region in a real quantum field theory is not that object. Local algebras are type III von Neumann factors, with the most singular subtype identified for physically reasonable theories [2, 4]. A type III factor admits no trace, so it has no density matrices, no partial trace, and no von Neumann entropy of a restricted state. The mathematical objects the earlier paper computed with do not exist there. What does exist is relative entropy, defined for states of an arbitrary von Neumann algebra through the relative modular operator [1], together with the data-processing inequality in full generality [7] and the characterization of its equality case [8]. Field-theory computations confirm that relative entropies of physical states are free of the ultraviolet

divergences that destroy the absolute quantities [9].

This paper asks which of the campaign's quantum objects survive the passage from finite models to that setting, and it answers by measurement rather than by declaration. The candidate thesis, fixed before any run, was that consumer-relative relative quantities survive while absolute quantities do not. Restricted relative entropies, their differences and orderings, task distortions, and the flip are built from expectation values and relative modular data, so each should have a well-defined limit as the model grows. The reduced density matrix, the entanglement entropy of a cut, and dephasing in a preferred basis are type I artifacts with no counterpart, so they should diverge or fail to be definable, and the campaign's sealed claims should turn out never to have depended on them. Either half of this thesis could have failed per quantity, and a failure would have been a recorded result.

Three instruments carry the answer. The first computes the campaign's sealed divergences by two independent type-III-native routes, one through the relative modular operator and one through states treated purely as expectation functionals on the consumer algebra, with no reduced density matrix formed anywhere in the code path, and requires exact agreement with the sealed finite instrument. The second runs a ladder of growing chains at fixed physics and measures which quantities converge and which diverge, with the entanglement entropy of the cut carried deliberately as the diverging control. The third reruns the flip protocol at every rung of the ladder under a budget accounting chosen, before the run, to be the one that can possess a limit at all.

Three claim types are kept separate throughout, exactly as in the earlier papers. Proved statements are the literature's theorems, cited and never retested here. Measured-in-model statements are facts about the de-

* andrew.bond@sjsu.edu; ORCID 0009-0003-2599-6158

clared finite models and their internal limits, and every number in this paper is of that kind at exploratory evidence label. Design statements describe instruments not yet run. One discipline is repeated from the earlier paper because it governs this one equally. Renaming known operator-algebra facts adds nothing. The existence of Araki relative entropy, the generality of data processing, and the type classification of local algebras are known mathematics, and calling them a bridge would be a re-labeling. The contribution of this paper, if any, is the measured bridge, meaning the quantitative demonstration that the campaign's specific sealed numbers are computed by the surviving objects and are already converged in the growing-model limit. The causal caution of the campaign applies verbatim. The consumer does not create anything physical by choosing what to observe, and the only admissible causal direction runs from the physical state through the accessible channel to the consumer-relative divergence.

II. THE HONESTY TABLE

Table I states, for each object the quantum track used, what it was in the finite models, what its status is when the algebra is type III, and what form the object must take for its claims to survive the passage. The table condenses a standing track document, and its verdicts rest on the cited literature, principally the modern review [2], the founding definition of relative entropy for general von Neumann algebras [1], the general data-processing inequality [7] and its equality case [8], the type III₁ identification for local algebras [4], the theory of standard split inclusions [5], the nuclearity condition that supplies them [6], the algebraic framework as a whole [3], and finiteness of relative entropies in conformal field theory as context [9].

Two reading rules govern the table. A verdict of no counterpart is not a defect of the quantum track. It is the reason the sealed claims were phrased through expectation values and divergences wherever possible. And the table is a standing document, so any run that contradicts a cell overwrites the cell with the change logged.

III. THE MODULAR DICTIONARY AND THE TB-0 INSTRUMENT

A. The surviving formula

For a faithful state ρ on a finite-dimensional algebra with eigendecomposition $\rho = \sum_i p_i |e_i\rangle\langle e_i|$, the vector $\Omega_\rho = \sum_i \sqrt{p_i} e_i \otimes e_i^*$ on the doubled space represents ρ , and under the identification of the doubled space with Hilbert-Schmidt operators Ω_ρ is the matrix $\sqrt{\rho}$. The relative modular operator of the pair (ρ, σ) acts on Hilbert-

Schmidt operators as

$$\Delta_{\sigma|\rho}(X) = \sigma X \rho^{-1}, \quad (1)$$

with spectrum the set of eigenvalue ratios s_i/p_j . Araki's formula [1] is

$$S(\rho \| \sigma) = -(\Omega_\rho, \ln \Delta_{\sigma|\rho} \Omega_\rho), \quad (2)$$

and in type I this equals the Umegaki trace formula identically, which a two-line computation in the joint eigenbasis confirms. Equation (2) contains no trace and no density matrix in its general form, only two states of a von Neumann algebra and the modular data they induce, which is why it is the formula that still parses when the algebra is type III.

One hazard is recorded because it bites in practice. References disagree on the index order of the relative modular operator, and some write the operator of Eq. (1) with the roles of the two states exchanged. Relative entropy is asymmetric, so a convention slip silently computes the divergence in the reversed order. The instrument pins its convention with an asymmetric commuting control pair whose divergences in the two orders differ in closed form, so a slipped convention fails the control rather than passing unnoticed.

A second route removes the density matrix entirely. For two states given only as expectation functionals on a finite-dimensional consumer algebra, the GNS construction from the reference functional yields a Hilbert space, a cyclic vector, and a relative modular operator built from the two functionals alone, and Eq. (2) then yields the restricted relative entropy without any reduced density matrix being formed at any point in the code path. In type I this must equal the restricted Umegaki value computed through the partial trace. The moral is stated once. The relative modular operator is the object that survives the passage, and density matrices are its type I shadow, useful for computation and meaningless as a foundation.

B. TB-0 results

The TB-0 instrument computes the campaign's sealed model, an eight-qubit transverse-field Ising chain with thermal reference, by three routes. Route one is the sealed spectral Umegaki instrument, untouched. Route two is Eq. (2) evaluated spectrally through the ratio set and the overlap matrix. Route three is the states-as-functionals GNS route on each consumer algebra, which also computes Uhlmann fidelity in GNS form against the trace-norm formula. The declared falsification bar was route disagreement above 10^{-10} on any state pair or region.

The measured worst disagreement over the entire grid is 9.4×10^{-16} , six decades inside the declared bar. The agreement holds across the closed-form checks, which cover the commuting Kullback-Leibler case at 5.6×10^{-17} , a de-phased pair at 9.4×10^{-16} , the global pair at 1.1×10^{-16} ,

TABLE I. The honesty table. One row per quantum-track object, condensed from the track document. Survives verbatim means the type I definition never used a density matrix and needs no rephrasing. TB-0, TB-1, TB-2, and TB-3 name the instruments of Secs. III through VII.

Object	Type I definition as used	Type III status	Bridge
Reduced density matrix	Partial trace of the global state, the state the consumer holds	No counterpart. A type III factor has no trace and no density operators [2]	The state as an expectation functional restricted to the algebra. The functional survives, the matrix was its type I shadow
Partial trace channel	CPTP map onto the accessible factor, the canonical consumer channel	No counterpart, there is no trace to take	Algebra restriction of the functional. Conditional expectations exist only in special cases and are flagged when used
Umegaki relative entropy	Spectral trace formula, the sealed QO-0 instrument	Counterpart exists, the Araki relative entropy through the relative modular operator [1]	Compute both routes at finite size and require exact agreement. TB-0
Data-processing inequality	Divergence non-increase under every channel, 32 sealed margins	Holds in full generality [7], equality characterized by sufficiency [8]	Survives verbatim. The sealed exact-sufficiency anchor becomes the Petz cocycle condition
Consumer equivalence	Equality of all expectation values on the consumer algebra	Survives verbatim, the definition never mentioned density matrices	None needed
Pinching channel	Diagonal-part map in the computational basis	No canonical counterpart, the map is basis dependent and no preferred maximal abelian subalgebra is supplied	None. Every statement needing it is flagged finite-size. The classical anchors live inside the commuting embedding, which is honestly type I
Entanglement entropy of the cut	Von Neumann entropy of the reduced state	Undefined, no von Neumann entropy of a region in type III [2]	None, and none needed. No sealed claim used it. TB-1 carries it as the deliberately diverging control
Uhlmann fidelity	Trace-norm formula, the flip's fidelity witness	Survives, defined on any von Neumann algebra through standard-form vector representatives	GNS-form computation checked against the trace-norm formula, folded into TB-0
The flip	Task win plus fidelity loss at matched budget, budgets as kept-qubit counts	Both witnesses are expectation and fidelity objects and survive. The kept-subset encoder family is type I language	Encoders rephrased as channels with a fixed absolute budget. Persistence measured on the ladder. TB-2
Consumer-family quantifier	Joint spectral-law requirement over nested windows, active with a shared event	Wedge families sharing an event are exactly the structure field theory possesses [2]. The lattice spectral law has no direct counterpart	Reformulation through modular flow. Design only. TB-3

and a near-degenerate pair with eigenvalue splittings swept downward, stable at 5.6×10^{-17} . It holds across the eight-qubit consumer sweep, where the single-site, pair, and half-chain consumers show functional route agreement with the partial-trace route at or below 2.3×10^{-16} . This result is deliberately a null test of mathematics, exactly as the sealed data-processing margins of the earlier paper were. Its job is to certify that the implementation computes the objects the theorems are about, because every later section leans on the modular route. The caveat is adjacent. Machine-precision agreement in type I does not test any type III statement, it certifies the instrument that the ladder then uses.

IV. THE SPLIT PROPERTY AS THE PHYSICAL REFERENT

Finite models cannot be honest stand-ins for local algebras, and the split property says what they can be honest stand-ins for. In a field theory satisfying the nuclearity condition on the energy-level density [6], for a double cone and a slightly larger concentric double cone there is an interpolating type I factor between the two local algebras,

$$\mathcal{A}(\mathcal{O}) \subset \mathcal{N} \subset \mathcal{A}(\widehat{\mathcal{O}}), \quad (3)$$

a standard split inclusion in the sense of Doplicher and Longo [5]. The interpolating factor is where density matrices, partial traces, and all of the campaign's finite machinery legitimately live. The local algebras on either side of it are type III₁ [4] and have none of that machinery.

The campaign adopts this as a standing rule. Every finite-size claim is a statement about a split-inclusion approximation, never about a local algebra. The collar, the gap between the region and its enlargement, is the resolution parameter of the approximation. On a chain the collar is concrete, since the consumer keeps a region and the enlarged region adds buffer sites on each side, and the collar width in sites plays the role the resolution parameter played in the campaign's classical projection-entropy track. The type III divergences are expected to reappear as the collar shrinks for absolute quantities and not for relative ones, which is the thesis of Sec. I restated in collar language. The ladder of the next section grows the system at fixed cut fraction, and every one of its witnesses is a witness about a family of such approximations.

V. THE SCALING LADDER

A. Design

The ladder runs chains of length $N = 4$ through 12 at fixed physics, coupling $J = 1$, transverse field $h = 2$ in the gapped paramagnetic regime, inverse temperature $\beta = 0.5$, and a diagonal excitation of strength $\theta = 0.8$ at the site adjacent to the half-chain cut. The consumers are the single boundary site, the boundary pair, and the kept half chain, the last being the consumer whose algebra grows with N . Two lists were declared before the run. The candidate-convergent list holds the consumer-relative restricted relative entropies of excitation against reference. The candidate-divergent list holds the von Neumann entropy of the thermal reference reduced to the half chain, included deliberately as the diverging control. If the control had failed to diverge, the ladder would have had no contrast and would have supported no convergence claim for anything.

B. Results

Every consumer-relative restricted relative entropy converges exponentially. Successive differences shrink roughly fifty-fold per rung. The site consumer moves from 1.2×10^{-3} at the first rung gap to 8.0×10^{-9} at the last, the pair consumer from 9.5×10^{-4} to 5.7×10^{-9} , and the half-chain consumer, whose algebra grows with the system, from 5.2×10^{-4} to 3.4×10^{-9} , reaching nine-digit stability by $N = 10$. The effective limits are

$$\begin{aligned} D_{\text{site}} &= 0.631800, & D_{\text{pair}} &= 0.674747, \\ D_{\text{half}} &= 0.675183, & D_{\text{global}} &= 0.723532, \end{aligned} \quad (4)$$

in nats, with the global divergence computed through $N = 10$. The Araki route of Sec. III agreed with the sealed Umegaki instrument at every rung and consumer within the 10^{-10} bar, and the states-as-functionals route agreed through $N = 8$, the largest size at which it was run.

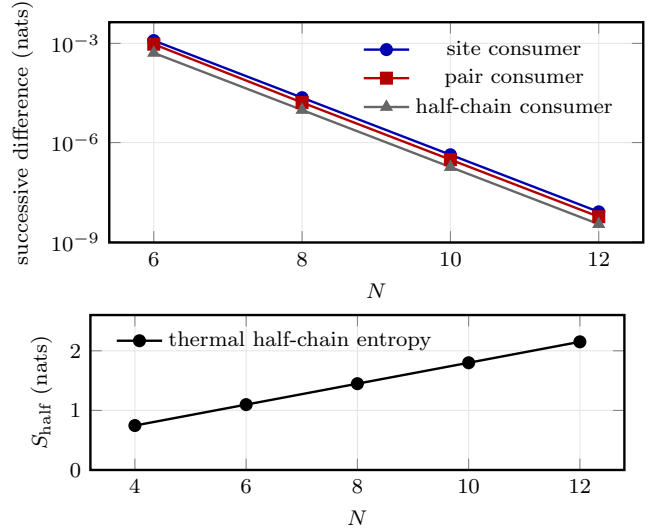


FIG. 1. The contrast pair of the ladder. Top, successive differences of the consumer-relative restricted relative entropies on a logarithmic axis, shrinking roughly fifty-fold per rung for all three consumers, including the half-chain consumer whose algebra grows with N . Bottom, the deliberately included absolute control, the von Neumann entropy of the thermal half chain, exactly extensive at 0.1756 nats per added site pair. Relative quantities have limits, the absolute entropy does not.

The diverging control diverged exactly as predicted. The thermal half-chain von Neumann entropy takes the values 0.746, 1.097, 1.448, 1.799, and 2.151 nats across the ladder, exactly extensive at 0.1756 nats per added site pair. Figure 1 shows the contrast pair, the plateaus of the relative quantities against the linear growth of the absolute one. One quantity, one verdict. The relative quantities have limits and the absolute entropy does not, which is the first quantitative confirmation of the thesis of Sec. I.

A corollary is stated plainly because it retroactively strengthens the campaign. The sealed quantum results were computed at $N = 8$, and Eq. (4) shows the $N = 8$ values already agree with the extrapolated limits to the precision at which they were quoted. The campaign's sealed quantum numbers are therefore effectively infinite-volume statements. The caveat is adjacent. This holds in the measured regime of a fixed physical excitation on a gapped reference at fixed cut fraction, the result carries the exploratory label, and the open questions about scaling regime and uniformity in Sec. VII are untouched by it.

VI. FLIP PERSISTENCE AT FIXED BUDGET

A. The budget decision, made before the run

The earlier paper's flip was measured at budgets counted in stored qubits, and a ladder forces a choice

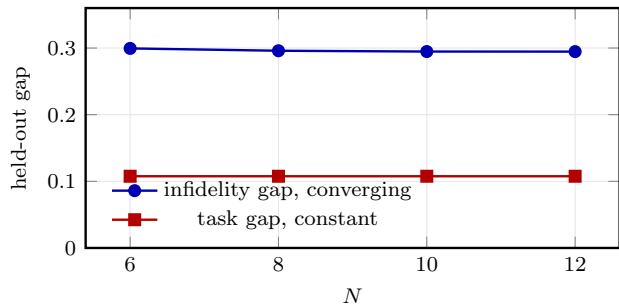


FIG. 2. Flip persistence at fixed absolute budget $k = 1$ across the ladder. The held-out task gap between the arms is 0.10762 at every rung, constant to 5.3×10^{-10} . The infidelity gap converges from 0.2994 to 0.2946 with successive differences shrinking roughly tenfold per rung. The flip holds at every N with the anti-arm worst on the task everywhere.

between two accountings, a fixed absolute number of stored qubits and a fixed fraction of the kept region. The choice was made before the run and recorded as binding, with fixed absolute budget primary, for two reasons. The task is local and its relevant halo converges by the ladder result of Sec. V, so scarcity relative to the task is a notion independent of system size. And a fractional budget has no type III meaning at all, because algebra dimension is the quintessential type I artifact, so only the fixed-count object can possess a limit. The fraction was retained as a secondary diagnostic.

B. Results

The protocol reruns the sealed flip design, with its training and held-out scenario grids, its declared win margins, and its mandatory anti-arm, on the kept halves of chains $N = 6$ through 12 . At budget $k = 1$ the flip holds at every rung, with the task arm winning the task, losing on fidelity, and the anti-arm worst on the task everywhere, which keeps every rung interpretable under the sealed rule. The held-out task gap between the arms is 0.10762 at all four rungs, constant to 5.3×10^{-10} by $N = 12$. The task functional, the task arm's encoder, and the fidelity arm's miss are all local objects, so the task gap is size-independent almost immediately. The infidelity gap converges from 0.2994 to 0.2946 with successive differences shrinking through 3.5×10^{-3} , 1.2×10^{-3} , and 1.1×10^{-4} . Figure 2 shows both gaps. At fixed absolute budget the flip is a thermodynamic-limit phenomenon, which is what the budget decision predicted and what a type III phrasing requires.

The secondary diagnostic did not trivialize, and the finding is reported adjacent to the result it qualifies. The flip region in budget count ran $\{1\}$, $\{1, 2\}$, $\{1\}$, and $\{1, 3\}$ across the four rungs, with collapse fractions 0.33 , 0.5 , 0.2 , and 0.5 , neither shrinking nor stabilizing. The cause is visible in the search records. The fidelity-optimal subset at intermediate budgets sometimes contains the task site

and sometimes does not, depending on which subsets best capture the reference correlations at each region size. The upper edge of the flip region is therefore a discrete subset-selection artifact at small region sizes, the clean convergent object is the fixed-count flip, and the fractional diagnostic is untestable below substantially larger regions. This is recorded as a finding rather than massaged away.

VII. WHAT THE BRIDGE DOES NOT DO

The bridge measured here is deliberately narrow, and its edges are stated so that no reader extends it.

The chain has no causal structure. There are no light cones, no Killing flow, and no horizon generators, and the collar of Sec. IV is a distance in sites, not a causal separation. No entropy-area statement appears anywhere in this paper, and the diverging control of Sec. V must not be read as one. No claim is made that any extrapolated limit in Eq. (4) equals a value in any quantum field theory. Convergence of a ladder of finite models is not existence of a continuum limit, and the finite models stand in for split-inclusion interpolators, never for local algebras.

One limitation is promoted to the standing open question of the track. The ladder grows the system at fixed cut fraction, which probes an infrared limit. The type III pathologies of the continuum are ultraviolet, arising as the collar of a split inclusion shrinks at fixed physical size. The ladder as run may therefore converge for reasons that say nothing about the ultraviolet passage, and a collar-width sweep at fixed physics is the declared candidate instrument for separating the two regimes. Until that sweep exists, the bridge is an infrared bridge.

The consumer-family quantifier remains at the design stage. Its type III reformulation replaces nested lattice windows with a family of split inclusions sharing an excitation region, replaces the lattice spectral law with a quadratic form built from the reference state's modular data, and inherits a circularity audit that the declared form must pass before any run. Nothing in this paper advances that instrument beyond design, and its honest possible endpoint includes termination at the design stage.

VIII. CONCLUSION

This paper measured a bridge between the campaign's finite quantum models and the operator-algebraic setting in which quantum field theory actually lives. The type-III-native routes, through the relative modular operator and through states as expectation functionals with no partial-trace primitive anywhere in the code path, reproduce the sealed instrument at 9.4×10^{-16} , six decades inside the declared bar. The ladder shows every consumer-relative restricted relative entropy converging exponentially to the limits of Eq. (4) while the entanglement entropy of the cut, the quantity outsiders most expect the program to

depend on, diverges exactly extensively as the deliberate control. The flip persists at every rung at fixed absolute budget with a task gap constant to nine decimals, under a budget accounting chosen before the run because it is the only one with a type III meaning. The corollary stands that the campaign’s sealed quantum results were already infinite-volume statements to their quoted precision.

The relabeling clause applies to everything else. The theorems cited here belong to Araki, Uhlmann, Petz, Fredenhagen, Doplicher and Longo, and Buchholz and Wichmann, and restating them adds nothing. Whatever in this paper merely renames known operator-algebra facts is not a contribution. The contribution, if any, is the measured bridge, the demonstration that the campaign’s

specific sealed numbers are carried by the objects that survive the passage and are already converged in the growing-model limit, together with the honest boundary of Sec. VII. Every result carries the exploratory label. Nothing in this paper claims anything about physical gravity, real horizons, or the actual values of any quantum field theory. The consumer does not create anything physical by choosing what to observe.

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